

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 May/June 2025 Worked Solutions

May/June 2025 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 May/June 2025

May/June 2025 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Binomial Expansion

- 1: (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(1 - 4x)^7$$

giving each term in simplest form.

(4)

In the series expansion of

$$(5 + kx)(1 - 4x)^7 \quad \text{where } k \text{ is a constant}$$

the coefficient of the term in x^2 is 1316

- (b) Use the answer to part (a) to find the value of k .

(2)

(Total for Question 1 is 6 marks)

(Total for Question 1 is 6 marks)

Topic group

- + - (b) Sets "336" 5 "28" k \times - = 1316 () 13 k =

Question 2

Circles

2: The line joining the points $(-2, 5)$ and $(4, 15)$ is the diameter of a circle C .

(a) Find an equation for C .

(5)

(b) Hence find the exact coordinates of the point on C that is nearest the x -axis.

(2)

(Total for Question 2 is 7 marks)

(Total for Question 2 is 7 marks)

Worked Solution - Question 2

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 3

Integration

3:

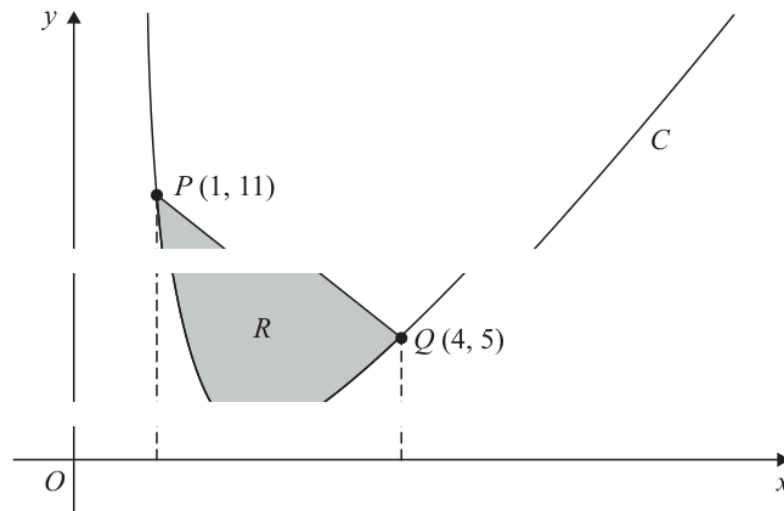


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve C with equation

$$y = 3x + \frac{16}{x^2} - 8 \quad x > 0$$

The points $P(1, 11)$ and $Q(4, 5)$ lie on C and are shown in Figure 1.

The region R , shown shaded in Figure 1, is bounded by C and line segment PQ .

Use algebraic integration to find the area of R .

(5)

(Total for Question 3 is 5 marks)

(Total for Question 3 is 5 marks)

Worked Solution - Question 3

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 4

Polynomials

4: The function $f(x)$ is defined by

$$f(x) = ax^3 + bx^2 + 5x - 3$$

where a and b are constants.

Given that $(x + 3)$ is a factor of $f(x)$,

(a) show that

$$b = 3a + 2$$

(2)

Given further that when $f(x)$ is divided by $(2x - 1)$ the remainder is $\frac{7}{4}$

(b) find the value of a and the value of b .

(4)

(c) Using algebra, find the quotient and the remainder when $f(x)$ is divided by $(x - 2)$

(3)

(Total for Question 4 is 9 marks)

(Total for Question 4 is 9 marks)

Worked Solution - Question 4

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: $+ + + -$ (Quotient $=$) $2 2$
 $12 29 x x + +$ Remainder $= 55$

Final answer

$+ + + -$ (Quotient $=$) $2 2 12 29 x x + +$ Remainder $= 55$

Question 5

Integration

5:

x	-2	-0.5	1	2.5	4	5.5	7
y	12	4.243	1.5	0.530	0.188	0.066	0.023

The table above shows corresponding values of x and y for

$$y = 3\left(\frac{1}{2}\right)^x$$

The values of y are given to 3 decimal places as appropriate.

- (a) Using the trapezium rule with all the values of y in the given table, obtain an estimate for

$$\int_{-2}^7 3\left(\frac{1}{2}\right)^x dx$$

giving the answer to one decimal place.

(3)

Using the answer to part (a) and making your method clear, estimate

(b) (i) $\int_{-2}^7 3\left(\frac{1}{2}\right)^{x+2} dx$

(ii) $\int_{-2}^7 (2^{-x} + 2x) dx$

(3)

(Total for Question 5 is 6 marks)

(Total for Question 5 is 6 marks)

Worked Solution - Question 5

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 6

Laws of Logarithms

6:

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

(i) Solve

$$2\log_2(4-x) = 3 + \log_2\left(\frac{x+11}{2}\right) \quad (5)$$

(ii) The curves C_1 and C_2 with equations

$$y = 3^{2x+1} \quad \text{and} \quad y = 6 \times 3^x$$

meet at the point P .

Find the exact coordinates of P , writing your answer in the form $(\log_3 a, b)$
where a and b are integers.

(5)

(Total for Question 6 is 10 marks)

(Total for Question 6 is 10 marks)

Worked Solution - Question 6

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 7

Geometric Sequences

7: In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) A **geometric** series begins

$$10 + 8 + 6.4 + \dots$$

- (a) Find the sum to infinity of this series.

(2)

Given that the k th term of this series is less than 0.0005

- (b) use algebra to find the smallest possible value of k .

(3)

- (ii) An **arithmetic** series begins

$$850 + 843 + 836 + \dots$$

Given that the sum of the first n terms of this series is S_n
find the greatest possible value of S_n

(4)

(Total for Question 7 is 9 marks)

(Total for Question 7 is 9 marks)

Worked Solution - Question 7

1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

3. State the final result

Write the answer in the form requested by the question: - = States or implies that
 $122 n = 122 \cdot 2 \cdot 850 \cdot 121 \cdot 7 \cdot 52033 \cdot 2$ $S = \times + \times - =$

Final answer

- = States or implies that $122 n = 122 \cdot 2 \cdot 850 \cdot 121 \cdot 7 \cdot 52033 \cdot 2$ $S = \times + \times - =$

Question 8

Trigonometric Equations

8: In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < \pi$, the equation

$$3 \tan \left(2x + \frac{\pi}{5} \right) = \sqrt{3}$$

giving the answers in radians in the form $k\pi$, where k is a rational constant to be found.

(3)

- (ii) Solve, for $0 \leq \theta < 360^\circ$, the equation

$$5 \sin \theta \tan \theta = \cos \theta + 4$$

giving your answers, in degrees, to one decimal place.

(5)

(Total for Question 8 is 8 marks)

(Total for Question 8 is 8 marks)

Worked Solution - Question 8

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

WMA12/01 MAY/JUNE 2025

11 marks

Question 9

Integration

9:

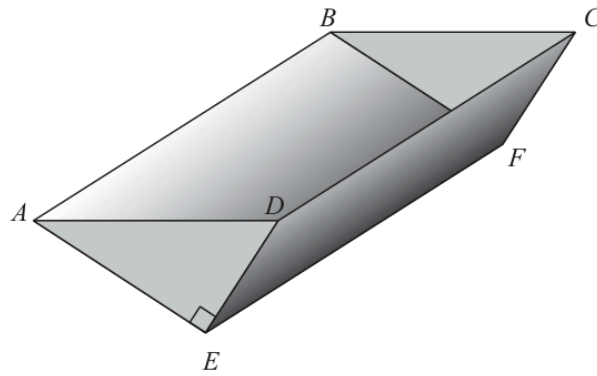


Figure 2

Figure 2 shows a sketch of an open container.

The sides $CDEF$ and $ABFE$ are rectangles.

The ends ADE and BCF are congruent (identical) right-angled triangles.

The container is made from metal of negligible thickness.

Given that

- $AE = BF = 3x$ metres
- $DE = CF = 2x$ metres
- $AB = DC = EF = L$ metres

and the capacity of the container is 12 m^3

(a) show that the area of metal used to make the container, $S \text{ m}^2$, is given by

$$S = Px^2 + \frac{Q}{x}$$

where P and Q are positive integers to be found.

(4)

Given that x can vary,

(b) use algebraic calculus to find the minimum value of S , giving your answer to one decimal place.

(5)

(c) Justify that the value of S found in part (b) is a minimum.

(2)

(Total for Question 9 is 11 marks)

(Total for Question 9 is 11 marks)

Worked Solution - Question 9

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $\frac{1}{2} \ln 2$, hence minimum $\frac{1}{2} \ln 2$

Final answer

$\frac{1}{2} \ln 2$, hence minimum $\frac{1}{2} \ln 2$

Question 10

Proof

10:

In this question you must show detailed reasoning.

Use algebra to prove by exhaustion that,

for all positive integers m that are **not** multiples of 3, the value of

$$m^2 + 3m + 2$$

is always a multiple of 3

(4)

(Total for Question 10 is 4 marks)

(Total for Question 10 is 4 marks)

Worked Solution - Question 10

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: $+++=++=++$ And gives a minimal conclusion $**$

Final answer

$+++=++=++$ And gives a minimal conclusion $**$